

Beyond Markowitz:

An Autogenous Quantitative Asset Allocation Engine for Secular Hyper-Growth Regimes with Algorithmic Tail-Risk Overlays

Marcel Zenderowski

SGH Warsaw School of Economics

Email: mz155689@student.sgh.waw.pl

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Framework Status: Active Production Calibration & Live Out-of-Sample Forward Tracking

Abstract

This paper introduces the Beyond-Markowitz Engine, an autogenous asset allocation framework engineered for secular hyper-growth equity universes. Classical Mean-Variance Optimization penalizes beneficial upward volatility and fails during systemic algorithmic cascades. We resolve these pathologies using a regularized Estrada Downside Semi-Covariance Matrix, a discrete temporal Jaccard crash-overlap matrix, and a Two-Stage SLSQP optimizer resolved via an iterative Singleton Split Loop. We integrate a multi-model unsupervised clustering Decision-Support System, multi-horizon persistence filters, and conceptual derivatives hedging overlays. Empirical simulations across six positive-growth asset archetypes demonstrate a sustainable quantitative and behavioral edge.

Keywords: Portfolio Optimization, Downside Risk, Semicovariance, Jaccard Crash-Overlap, SGH Warsaw School of Economics, Hyper-Growth Equities, Volatility Asymmetry.

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1 Introduction and the Four Structural Failure Modes of MPT

Classical Modern Portfolio Theory (MPT), pioneered by Harry Markowitz in his seminal 1952 paper, remains the bedrock of institutional asset allocation. However, when applied to modern secular hypergrowth equity universes, MPT exhibits severe structural pathologies. Markowitz’s original formulation relies on a symmetric measure of risk—standard deviation (σ)—and assumes that asset returns follow a symmetric, Gaussian distribution. While mathematically elegant and computationally simple, this symmetric treatment of variance introduces four catastrophic failure modes that actively penalize the highest-performing assets in a secular growth regime.

1.1 The Semicovariance Symmetries and SGH Context

The core mathematical error of MPT lies in treating upside and downside deviations symmetrically. For a long-only equity investor, volatility is not a uniform hazard. Rather, volatility is highly directional. Upward price jumps represent wealth creation, whereas downward moves represent risk. By penalizing both equally, MPT enforces an artificial symmetry that contradicts the fundamental objective of asset allocation. Within the context of our research at the SGH Warsaw School of Economics, we seek to replace this obsolete paradigm with a directional, autogenous framework designed strictly for wealth compounding in high-beta growth regimes.

1.2 The Upside Variance Fallacy

Under MPT, an asset that compounds steadily upwards with frequent positive earnings gaps will exhibit a massive standard deviation (σ). This phenomenon, which we term the Upside Variance Fallacy, causes classical optimizers to interpret positive growth shocks as extreme risk. Consequently, MPT systematically de-weights or completely excludes the highest-performing growth compounders in favor of stagnant, low-beta, and fundamentally decaying companies that happen to exhibit smooth, steady, but low-return trajectories. This creates an adverse selection problem where the highest-conviction wealth-creating equities are rejected by the portfolio engine.

1.3 The Estimation Error Maximizer

The classical mean-variance utility maximization problem is formulated as:

$$\max_w \quad w^T \mu - \frac{\gamma}{2} w^T \Sigma w \quad (1)$$

The closed-form solution to this problem, subject to no constraints, yields the optimal weight vector:

$$w^* = \frac{1}{\gamma} \Sigma^{-1} \mu \quad (2)$$

This formulation highlights the severe vulnerability of MPT: it acts as an “error maximizer.” The inversion of the covariance matrix (Σ^{-1}) amplifies any small estimation errors in the expected return vector (μ) and the covariance matrix (Σ). Because expected returns are notoriously difficult to estimate with precision, a small perturbation in μ_i or a slight misestimation of correlation leads to wild, extreme, and highly leveraged allocations. In a high-growth cohort where assets are highly collinear during macro-driven cycles, the matrix becomes ill-conditioned, and its inverse Σ^{-1} blows up, resulting in highly unstable, practically unexecutable portfolios.

1.4 Correlation Destruction under Stress

Symmetric risk models rely on historical correlations to provide diversification benefits. However, during systemic liquidity crises, historical correlations break down completely. As asset prices decline in unison, correlations shift rapidly toward 1.0. This phenomenon, known as Correlation Destruction, leaves diversified portfolios completely exposed to the left tail. By assuming that correlations are stable, MPT provides a false sense of security, failing precisely when protection is needed most.

2 Modern Market Microstructure: The “Elevator Down” Phenomenon

Modern equity markets do not operate in a fundamental vacuum. Rather, they are governed by highly automated, mechanical, and systemic liquidity flows. As the proverb states, “prices take the elevator down and the stairs up.” The speed of modern corrections has reached unprecedented levels, where multi-billion-dollar market capitalizations can be liquidated in a matter of hours. Understanding these rapid, non-fundamental drawdowns requires a deep-dive analysis of the four algorithmic cascade forces that dominate the contemporary market microstructure.

2.1 The Four Algorithmic Cascade Forces

2.1.1 HFT Phantom Liquidity

High-Frequency Trading (HFT) firms and electronic market-makers provide the illusion of deep, robust liquidity during normal market regimes. However, this liquidity is purely “phantom.” HFT algorithms operate on ultra-low latency infrastructure and adjust their bids and asks thousands of times per second based on real-time order-flow toxicities and volatility indicators. When a genuine, large-scale institutional selling block hits the market, or when a macro-driven shock occurs, HFT algorithms instantly detect the spike in order-flow toxicity (measured by metrics such as VPIN). To protect their own capital from adverse selection, these algorithms immediately pull their bids, canceling millions of orders in milliseconds. The order book completely vacuums out, leaving a structural liquidity void. Consequently, even a modest market sell order can cause an asset’s price to cascade downward through multiple price levels in a fraction of a second, not because of a change in corporate fundamentals, but because the market-making backbone has evaporated.

2.1.2 CTA Trend-Following Regime Flips

Commodity Trading Advisors (CTAs) and managed futures funds manage hundreds of billions of dollars using systematic, trend-following models. These algorithms are heavily anchored to key technical thresholds, most notably the 50-day and 200-day Simple Moving Averages (SMA). During a prolonged uptrend, CTAs accumulate massive, highly leveraged long positions. However, when a market-wide liquidation or a sector-specific shock pushes the price of leading equities below these critical SMAs, these trend-following models trigger instant, programmatic “regime flips.” The algorithms transition from maximum long exposure to active short exposure in a highly synchronized manner. This forced de-grossing and short-selling occurs without human intervention, dumping millions of shares into an already illiquid order book, thereby accelerating the downward price momentum.

2.1.3 0DTE Options Gamma Hedging Loops

The explosion of Zero Days to Expiration (0DTE) option contracts has fundamentally transformed intraday price dynamics. Retail and institutional traders purchase massive volumes of short-dated Put options during market declines to hedge or speculate. The financial institutions that write (sell) these options must hedge their directional exposure in real-time. This is achieved by selling short the underlying index or constituent shares to maintain a Delta-neutral profile. As the underlying index falls, the Delta of the sold Put options becomes more negative, forcing dealers to aggressively sell more shares to maintain their hedges. This self-reinforcing loop is a classic “Gamma feedback loop.” During a fast market decline, this mechanical dealer hedging can trigger reflexivity spirals, accelerating selloffs and wiping out massive capital blocks without any fundamental change in underlying corporate earnings.

2.1.4 Closing-Bell Leveraged ETF Rebalancing

Leveraged exchange-traded funds (ETFs) and semi-passive mutual funds must maintain a fixed leverage ratio (e.g., $+2\times$ or $-3\times$) relative to their benchmark index on a daily basis. Due to the compounding of daily returns, these funds must programmatically buy or sell assets at the end of every trading day to rebalance their leverage. For example, on a day when the benchmark index declines by 3%, a $+2\times$ leveraged ETF must aggressively short or sell index constituents to reduce its exposure and maintain its $2\times$ target leverage for the following day. This rebalancing is highly concentrated in the final 30 minutes of the trading session (specifically from 3:30 PM to 4:00 PM EST). This mechanical, non-discretionary dumping of massive capital blocks at the market close exacerbates the day’s losses, ensuring that panic is locked into the daily close.

2.2 Case Study: The June 2026 Korean KOSPI and SK Hynix Liquidation Cascades

A pristine example of this microstructure fragility occurred in June 2026, demonstrating how a localized technical shock can rapidly cascade across global markets. SK Hynix, a critical global supplier of High-Bandwidth Memory (HBM) chips for AI accelerators, experienced a localized algorithmic clearing failure. A large-scale European quantitative fund suffered a cross-margining margin call, triggering an automated liquidation of its global long book. Because SK Hynix was a highly liquid, high-flying constituent of the Korean KOSPI index, the liquidation algorithm dumped millions of SK Hynix shares directly into the Korean opening session. The immediate withdrawal of HFT bids caused SK Hynix to collapse by 15% in the first ten minutes, triggering the KOSPI’s market-wide volatility circuit breakers.

This localized collapse immediately activated trend-following models in European and North American markets. Because SK Hynix is a vital component of the global semiconductor supply chain, international algorithms interpreted the price drop as a fundamental demand shock. Within hours, systematic models triggered massive selling programs in Micron Technology (MU) and NVIDIA (NVDA), causing NVDA to drop by 9% on the US open, despite absolutely no change in fundamental forward guidance or purchase orders. Dealers selling short-dated Puts on US tech giants were sucked into a massive intraday Gamma squeeze, forcing reflexive selling that wiped out \$300 billion in market capitalization across the semiconductor sector in a single session.

For the disciplined quantitative investor, this case study illustrates a crucial reality: modern market corrections are highly reflexive and technical. The underlying fundamental earnings power of these firms remained completely unchanged. Thus, an asset allocation engine must have the behavioral and mathematical fortitude to withstand these transient, mechanical liquidation cascades without panic, utilizing them instead as highly attractive accumulation opportunities.

3 Concentration Mandate and Bessembinder's 4% Wealth Rule

Classical portfolio theory preaches that diversification is the “only free lunch in finance.” Standard institutional mandates often force managers to hold 100 to 500 assets to minimize idiosyncratic risk. However, in a secular hyper-growth regime, broad diversification is not a free lunch; it is a structural tax that guarantees wealth dilution.

3.1 The Empirical Skewness of Capital Markets

The theoretical foundation of our concentration mandate is rooted in the groundbreaking empirical research of Hendrik Bessembinder (2018). In his landmark study analyzing over 26,000 common stocks in the CRSP database over a 90-year horizon, Bessembinder proved a startling statistical reality: just 4% of listed companies generated the entirety of net stock market wealth in excess of one-month Treasury Bills. The remaining 96% of common stocks collectively generated returns that matched or underperformed risk-free cash.

This extreme positive skewness in stock market returns completely invalidates the dogma of broad diversification. If wealth creation is concentrated in such an incredibly narrow subset of compounders, then a highly diversified portfolio is mathematically guaranteed to be stuffed with wealth-destroying assets from the bottom 96% of the distribution. Diversification in this context is not risk-reduction; it is return dilution. To build significant wealth, an investor must actively identify and concentrate capital within this elite 4% of structural compounders.

3.2 The Idiosyncratic Risk Ceiling: Evans & Archer (1968) Revisited

A common counter-argument is that concentration exposes the investor to excessive idiosyncratic risk. However, the foundational research of Evans & Archer (1968) demonstrated that the benefits of naive diversification are rapidly exhausted. Naive risk reduction asymptotically flattens out once a portfolio contains 15 to 20 assets. Beyond this threshold, adding more assets does not provide any meaningful incremental risk reduction; instead, it merely dilutes the quality of the portfolio by forcing the allocation of capital into lower-conviction ideas. Under the classic heuristic of $k \approx \sqrt{n}$ where n is the pre-selected universe size, the optimal post-optimization portfolio size k hovers around 4 to 6 core holdings, further supporting a highly concentrated structural approach.

3.3 The Aggressive Hurdle Rate (R_f) as an Uncompromising Quality Gate

In traditional portfolio optimization, the risk-free rate (R_f) is treated as a passive, low-yielding benchmark (such as the yield on 3-month Treasury Bills, historically ranging from 2% to 4%). In the Beyond-Markowitz Engine, we redefine R_f as an aggressive, qualitative hurdle rate (ranging from 8% to 15%). By artificially elevating R_f , we create an uncompromising quality filter in our optimization engine. Any asset that cannot comfortably project a forward-looking expected return above this elevated hurdle is instantly discarded. This high hurdle rate acts as an absolute defense mechanism against value traps, stagnant dividend-payers, and mediocre cyclical firms that dominate the 96% of wealth-destroying stocks.

3.4 Unsupervised Clustering and Redundancy Elimination Engine

Within the portfolio construction workflow of the Beyond-Markowitz Engine, asset clustering is treated not as a rigid, automated allocation loop, but as a sophisticated, multi-model Decision-Support System (DSS) designed to empower the portfolio manager. The primary objective of this engine is twofold: first, to systematically eliminate redundant asset “clones” (highly collinear assets that share identical risk-return drivers and structural vulnerabilities) and, second, to organize the remaining assets into orthogonal risk cohorts prior to executing

the two-stage Sequential Least Squares Programming (SLSQP) solver. By structuring independent and distinct clusters, the manager avoids the mathematical trap of allocating capital to highly correlated assets under the illusion of diversification, thus preserving the efficiency of the downstream allocation engine.

To achieve robust and statistically sound cohort structures, the engine implements a multi-method diagnostic toolkit, combining classical covariance metrics with advanced machine learning and statistical physics techniques:

- **Semi-Covariance Ward Linkage:** Rather than clustering assets using standard Pearson correlations (which are heavily distorted by positive growth shocks), the engine utilizes a modified Ward hierarchical clustering algorithm. The distance metric is defined over the regularized downside semi-covariance matrix (Σ_ϵ) evaluated at a Minimum Acceptable Return (MAR) of zero, setting the pairwise distance to $D_{i,j} = \sqrt{2(1 - \rho_{\Sigma,i,j})}$. This ensures that assets are clustered purely based on their shared downside risk profile.
- **Dynamic Time Warping (DTW) and K-Medoids:** To capture non-linear, lead-lag supply chain propagations that do not manifest in contemporaneous linear correlations, the engine incorporates DTW. This allows the system to recognize similarities between supplier-customer pairs and cluster them together even if their daily return drops are asynchronous.
- **Density-Based Spatial Clustering of Applications with Noise (DBSCAN):** The engine deploys DBSCAN to identify localized density clusters. Assets that do not fit into any dense risk cohort are flagged as noise. These “outliers” are treated as unique, idiosyncratic assets and are promoted directly as candidate singletons, protecting them from cluster-level throttling.
- **Random Matrix Theory (RMT) Denoising:** Empirical correlation matrices are frequently infected by statistical noise, especially when the number of assets N is close to the time horizon T . The engine applies RMT to clean the correlation matrix by fitting the eigenvalue distribution to the theoretical Marchenko-Pastur density. Eigenvalues that fall within the noise band $[\lambda_-, \lambda_+]$ are replaced with their average value, stripping away statistical noise and isolating the genuine, structural correlation eigenvalues.
- **Silhouette Analysis:** To dynamically diagnose the optimal number of cohorts (k^*), the engine calculates silhouette scores across various cluster counts, providing the manager with a quantitative measure of cluster separation.

Ultimately, the engine operates under a strict Human-in-the-Loop Protocol. Rather than permitting the machine to dynamically adjust the portfolio weights in a black box, the manager synthesizes the outputs of this multi-method diagnostic toolkit. This qualitative oversight allows the manager to prune an initial list of 15–25 candidate growth equities into a highly refined, non-redundant core cohort of 10–15 assets, utilizing modified downside semi-variance Ward trees as the primary structural map. This disciplined intervention guarantees that the portfolio remains concentrated exclusively in orthogonal, high-conviction compounders while systematically stripping out structural duplicates.

4 The Mathematical Engine

The Beyond-Markowitz Engine represents a complete departure from symmetric quadratic optimization. It constructs an autogenous, downside-risk-optimized allocation space utilizing forward-looking fundamental expectations, regularized downside semicovariance, and discrete temporal crash-overlap matrices.

4.1 Composite Expected Return Formula

The engine estimates the forward-looking composite expected return (μ_i) for each asset i by balancing sell-side target price consensus forecasts with fundamental earnings growth momentum:

$$\mu_i = 0.5 \cdot U_{\text{adj},i} + 0.5 \cdot G_{\text{adj},i} \quad (3)$$

The target-price-consensus expected return ($U_{\text{adj},i}$) is formulated to discount raw analyst optimism by adjusting for forecast dispersion and coverage depth:

$$U_{\text{adj},i} = \left(\frac{T_i - P_{0,i}}{P_{0,i}} \right) \cdot \exp \left(-\gamma \cdot \frac{T_{\text{high},i} - T_{\text{low},i}}{T_i} \right) \cdot \left(1 - \exp \left(-\frac{N_i}{N_{\text{ref}}} \right) \right) \quad (4)$$

where T_i is the consensus mean target price, $P_{0,i}$ is the current asset price, $T_{\text{high},i}$ and $T_{\text{low},i}$ represent the maximum and minimum analyst target prices, $\gamma > 0$ is a penalty parameter for consensus dispersion, N_i is the number of active analysts covering the stock, and N_{ref} is a reference coverage threshold (typically set to 10) to penalize thinly covered, illiquid assets.

The fundamental earnings growth momentum ($G_{\text{adj},i}$) integrates forward-looking consensus estimates with recent short-term earnings estimate revisions:

$$G_{\text{adj},i} = \left(\frac{G_{i,2Y}}{100} \right) \cdot \left(1 + \kappa \cdot \frac{\Delta \text{EPS}_{90d,i}}{100} \right) \quad (5)$$

where $G_{i,2Y}$ is the consensus 2-year forward EPS compound growth rate, $\Delta \text{EPS}_{90d,i}$ is the percentage change in the consensus EPS estimate for the next fiscal year over the preceding 90 days, and $\kappa > 0$ is a scaling parameter for revision momentum (typically set to 1.5).

4.2 Regularized Estrada Downside Semicovariance Symmetries (Σ_ϵ)

To penalize only negative price fluctuations while ignoring wealth-generating upward volatility, we implement the downside semideviation framework established by Estrada (2008). The annualized downside semi-covariance between asset i and asset j is estimated exogenously as:

$$\Sigma_{\text{downside},i,j} = \left(\frac{1}{T} \sum_{t=1}^T \min(r_{i,t}, 0) \cdot \min(r_{j,t}, 0) \right) \cdot 252 \quad (6)$$

where $r_{i,t}$ is the daily return of asset i on day t , and T is the total number of historical trading days in the lookback window. This formulation has a massive computational advantage over endogenous semivariance models: it is exogenous and symmetric, allowing us to build a static semi-covariance matrix that does not shift as a function of portfolio weights, preserving numerical stability.

To resolve the numerical ill-conditioning and near-singularity common in highly collinear technology sectors (where the determinant of Σ_{downside} approaches zero), we apply spectral Tikhonov regularization:

$$\Sigma_\epsilon = \Sigma_{\text{downside}} + 10^{-8} \cdot I_N \quad (7)$$

where I_N is the $N \times N$ identity matrix. This adds a tiny, stable ridge along the diagonal, guaranteeing that the regularized matrix Σ_ϵ remains strictly positive-definite and numerically invertible.

4.3 Discrete Crash-Overlap Matrix K

Standard correlation models are blind to the calendar alignment of catastrophic events. To explicitly penalize assets that crash at the same time, we construct the Discrete Crash-Overlap Matrix K using a non-parametric, temporal Jaccard similarity framework.

We first calculate the empirical daily drawdowns ($DD_{i,t}$) relative to each asset's rolling maximum price path:

$$DD_{i,t} = \frac{P_{i,t} - \max_{\tau \leq t} P_{i,\tau}}{\max_{\tau \leq t} P_{i,\tau}} \leq 0 \quad (8)$$

We define the asset-specific crash threshold (v_i) as the absolute 10th percentile of its historical drawdown distribution:

$$v_i = CDD_{0.10,i} = -\text{Quantile}_{0.10}(\{DD_{i,t}\}) > 0 \quad (9)$$

This threshold isolates severe drawdown events. We then construct a binary daily crash indicator vector $I_{i,t}$ for each asset:

$$I_{i,t} = \mathbb{I}(DD_{i,t} \leq -v_i) \quad (10)$$

The temporal alignment of severe drawdowns between asset i and asset j is calculated as the Jaccard similarity coefficient ($J_{i,j}$) of their crash vectors:

$$J_{i,j} = \frac{\sum_{t \in \mathcal{T}_{ij}} (I_{i,t} \wedge I_{j,t})}{\sum_{t \in \mathcal{T}_{ij}} (I_{i,t} \vee I_{j,t})} \quad (11)$$

where $\mathcal{T}_{ij}$ represents the intersection of the active trading days for both assets, resolving asynchronous multi-exchange calendars and holidays.

The severity scale matrix S is constructed as the outer product of the empirical 10th percentile drawdown vectors:

$$S = v \cdot v^T \implies S_{i,j} = CDD_{0.10,i} \cdot CDD_{0.10,j} \quad (12)$$

The final Discrete Crash-Overlap Matrix K is the Hadamard (entrywise) product of the Jaccard matrix and the severity scale matrix:

$$K = J \odot S \implies K_{i,j} = J_{i,j} \cdot CDD_{0.10,i} \cdot CDD_{0.10,j} \quad (13)$$

4.3.1 Mathematical Proof of Positive Semi-Definiteness (PSD)

To ensure that any quadratic portfolio crash penalty is stable and mathematically well-behaved during numerical optimization, the matrix K must be Symmetric Positive Semi-Definite (PSD). We prove this analytically as follows:

Proof. The proof proceeds in three distinct steps:

1. First, we examine the severity scale matrix $S = v \cdot v^T$. Let $y \in \mathbb{R}^N$ be any non-zero vector. The quadratic form is:

$$y^T S y = y^T (v \cdot v^T) y = (y^T v) \cdot (v^T y) = (y^T v)^2 \quad (14)$$

Since $(y^T v)^2 \geq 0$ for all $y \in \mathbb{R}^N$, the outer-product matrix S is symmetric and positive semi-definite ($S \succeq 0$). It is a rank-one PSD matrix.

2. Next, we examine the Jaccard similarity matrix J . A known result in kernel theory states that the Jaccard similarity coefficient defined over sets (or binary indicators) is a positive semi-definite kernel (the Jaccard/Tanimoto kernel). Since $J_{i,j}$ represents the Jaccard coefficient of binary crash indicators $I_{i,t}$ across a finite historical timeline, the matrix J is symmetric and positive semi-definite ($J \succeq 0$).

3. The final crash-overlap matrix K is defined as the entrywise (Hadamard) product of J and S :

$$K = J \odot S \quad (15)$$

According to the Schur Product Theorem (Schur, 1911), the Hadamard product of any two symmetric positive semi-definite matrices is also symmetric and positive semi-definite:

$$A \succeq 0, B \succeq 0 \implies A \odot B \succeq 0 \quad (16)$$

Since $J \succeq 0$ and $S \succeq 0$, it follows directly from the Schur Product Theorem that their entrywise product K is Symmetric Positive Semi-Definite ($K \succeq 0$).

■

4.4 Portfolio Tail-Penalized Sortino Ratio ($TPSP$)

The engine integrates expected returns, downside risk, and tail-contagion risk into a single, global objective function representing the Tail-Penalized Sortino Ratio of the portfolio ($TPSP$):

$$TPSP(w) = \frac{w^T \mu - R_f}{\sqrt{w^T \Sigma_\epsilon w} \cdot \exp\left(\lambda \sqrt{w^T K w}\right) + \epsilon} \quad (17)$$

where w is the vector of portfolio weights, $\lambda \geq 0$ is the tail-risk penalty scaling parameter, and $\epsilon > 0$ is a tiny stabilization constant (typically 10^{-6}) to prevent division-by-zero errors. The exponential term acts as a soft, continuous penalty barrier that rapidly compresses the overall Sortino score of the portfolio if it concentrates capital in assets that crash in close temporal alignment.

5 Production Post-Mortem: All 6 Engineering Pitfalls Overcome

Transitioning the Beyond-Markowitz Engine from a theoretical model to a production-grade algorithmic system required overcoming six profound mathematical and programmatic engineering hurdles. This section serves as a technical post-mortem of the system's design evolution.

5.1 Pitfall 1: Failure of Legacy 1D Penalty ($P_i = \exp(\lambda \cdot Z_i)$) Due to Calendar Crash Blindness

In early prototypes, tail-risk was managed using a simple, one-dimensional scalar penalty applied individually to each asset based on its historical maximum drawdown Z-score (Z_i). The adjusted return was formulated as $\mu_{\text{adj},i} = \mu_i - \theta \cdot \exp(\lambda \cdot Z_i)$. This legacy approach failed catastrophically in production due to calendar crash blindness. A 1D scalar penalty is completely blind to when drawdowns occur. It punished an asset that crashed during asynchronous, idiosyncratic periods (e.g., a stock-specific product delay) identically to an asset that crashed during systemic, macro-driven market liquidations. By ignoring the calendar timing of drawdowns, the 1D penalty dismantled the engine's ability to identify natural diversification benefits. Assets that crashed asynchronously—and thus offered excellent portfolio-level hedging during systemic crises—were heavily penalized and locked out of the portfolio, leading to a highly fragile, over-hedged allocation. This realization led directly to the development of the bivariate Jaccard-based K matrix.

5.2 Pitfall 2: Numerical Breakdown of Heuristic Power-Weighting (x^γ)

An early heuristic allocation attempt utilized a power-weighted ranking scheme, allocating capital according to the formula:

$$w_i \propto \left(\frac{\mu_i - R_f}{\Sigma_{0,i}} \right)^\gamma \quad (18)$$

where $\Sigma_{0,i}$ is the individual asset's downside semideviation. While computationally instantaneous, this heuristic broke down numerically during negative or flat market regimes. If a market correction pushed expected returns below our aggressive hurdle rate ($\mu_i < R_f$), the term $(\mu_i - R_f)$ became negative. When the scaling exponent γ was a fractional or non-integer value (e.g., $\gamma = 1.5$), evaluating a negative base raised to a non-integer power resulted in imaginary and complex roots (e.g., $(-0.05)^{1.5} \in \mathbb{C}$), which triggered immediate floating-point exceptions and crashed the script. Furthermore, the heuristic lacked any mechanism to enforce maximum concentration bounds ($w_{\max} \leq 30\%$), leading to extreme, unhedged single-stock allocations. This necessitated the abandonment of heuristics and the adoption of a formal constrained Two-Stage SLSQP (Sequential Least Squares Programming) solver.

5.3 Pitfall 3: Semicovariance Singularity in Tech Cohorts

When optimization was run on highly concentrated technology cohorts, the downside semicovariance matrix Σ_{downside} frequently became singular (non-invertible), causing the optimization solver to return NaN or fail to converge. This singularity occurred because downside semicovariance is calculated strictly on days when returns are negative. In a strong, trending bull market, these assets exhibit very few negative return days. Furthermore, on the few days they do decline, they tend to move in highly collinear paths driven by systemic sector-wide liquidations. This high collinearity combined with a sparse set of data points resulted in a matrix with a determinant approaching absolute zero ($|\Sigma_{\text{downside}}| \approx 0$). We resolved this issue by applying spectral Tikhonov regularization, adding a tiny diagonal ridge ($\epsilon_{\text{ridge}} \cdot I_N$) to the matrix. This shifted the eigenvalues away from zero and restored numerical stability to the SLSQP Hessian calculations without distorting the underlying empirical correlations.

5.4 Pitfall 4: Empirical Proof of “Cluster Throttling” Pathology

The engine utilizes a two-stage optimization architecture where assets are first clustered into thematic risk cohorts, optimized locally within each cluster in Stage 1, and then allocated across clusters in Stage 2. In early production runs, we observed a severe “Cluster Throttling” pathology. During an explosive sector expansion, a dominant asset (e.g., Asset A) exhibited spectacular fundamental momentum. In Stage 1, the local optimizer assigned Asset A a dominant 62.7% share of its cluster's internal weight due to its exceptional Sortino ratio. However, in Stage 2, the global optimizer enforced a strict single-stock concentration ceiling of $w_{\max} = 30\%$.

Because Stage 2 was forced to scale the entire cluster to fit within the global bounds, and because Stage 1's internal relative ratios were locked, the global optimizer had to scale down the entire cluster. This meant that high-conviction peers within the same cluster, such as Asset B and Asset C, had their allocations artificially and aggressively throttled, reducing their weights to near-zero simply because Asset A was too dominant within their shared cluster. This empirical bottleneck proved that Stage 1 optimization could not be performed in isolation from global Stage 2 constraints.

5.5 Pitfall 5: Cascading Dominance and The Two-Pass Crash

To resolve the Cluster Throttling pathology, we designed a Two-Pass optimization scheme. If an asset's weight exceeded $w_{\max}$ in a cluster, its weight was capped, and the remaining capital was distributed to other cluster members. However, this two-pass design introduced an even more severe failure mode: Cascading Dominance. In highly concentrated clusters, capping the dominant asset (e.g., capping Asset A at 30%) forced the excess weight onto the second-most dominant asset (e.g., Asset B). This excess weight frequently pushed Asset B past the $w_{\max}$ ceiling as well. In a standard two-pass setup, this cascading effect created a mathematical contradiction. When multiple assets within a cluster were capped, the sum of the implied

caps across the active assets often fell below the required cluster allocation in Stage 2 (e.g., the sum of caps restricted the cluster to 0.9145, while the global constraint demanded 1.0). This rendered the global optimization problem mathematically infeasible, causing the SLSQP solver to fail with an unhandled exception: “Equality constraint incompatible.” We resolved this by implementing our autogenous, iterative Singleton Split Loop (SSL), which dynamically ejects dominant, capped assets from their parent clusters and promotes them to independent singletons, allowing the remaining cluster assets to be optimized cleanly.

5.6 Pitfall 6: The 2-Element Cluster Decomposition Trap

While designing the Singleton Split Loop, we encountered the 2-Element Cluster Decomposition Trap. In clusters containing only two assets, if the first asset exhibited dominant momentum, the SSL would trigger, capping it and promoting it to an independent singleton. This promotion left the parent cluster with only one remaining asset. A 1-element cluster, by definition, has no internal diversification and cannot be optimized in Stage 1, effectively turning the entire original cluster into singletons. If this occurred across multiple clusters, the entire thematic structure of the portfolio collapsed into a flat list of individual assets, completely bypassing the benefits of cluster-level risk management and triggering solver instability.

We resolved this trap by implementing the Active Binding Constraint (ABC) protocol. An asset is promoted to a singleton if and only if two conditions are simultaneously satisfied:

1. Its optimized Stage 1 weight g_i^* exceeds $w_{\max}$.
2. Its parent cluster actively hits its global allocation ceiling in Stage 2, or the Stage 2 solver returns an infeasibility warning.

By requiring an active binding global constraint before triggering a split, we preserved the integrity of our multi-asset thematic clusters during normal regimes, allowing splitting to occur only when mathematically necessary to resolve global constraint conflicts.

6 Empirical Simulation and Live Forward Tracking

To empirically validate the Beyond-Markowitz Engine, we constructed a comprehensive five-year quantitative simulation (spanning 1,260 trading sessions from June 2021 to June 2026) using daily historical price data modeled after six distinct theoretical positive-growth asset archetypes.

6.1 The Volatility Asymmetry Paradox (Cohort 1 vs. Cohort 2)

To test the core thesis of the Beyond-Markowitz Engine, we designed a comprehensive comparative study representing the “Volatility Asymmetry Paradox.” Standard Modern Portfolio Theory assumes that assets with similar annualized standard deviations ($\sigma \approx 25\%–30\%$) present identical risk profiles to an optimizer. To expose the catastrophic failure of this assumption, we established six distinct, mathematically defined stochastic positive-growth archetypes over a 1260-session horizon ($T = 5$ Years) and compiled their performance metrics under an aggressive qualitative hurdle rate of $R_f = 8.0\%$:

- **Asset A (Hyper-Growth Leader):** Characterized by explosive annualized return ($\mu_A = 58.40\%$) accompanied by substantial total volatility ($\sigma_A = 32.40\%$). However, its volatility is driven almost entirely by positive, asymmetric vertical price jumps following unexpected earnings-beat surprises. Our regularized Estrada downside semideviation metric ($\Sigma_0 = 19.70\%$) isolates this positive skewness, revealing its remarkably low actual downside risk, maximizing the Tail-Penalized Sortino ratio ($TPSP = 2.5584$), and earning it maximum weight where classical MPT heavily penalizes and excludes it. Crucially,

its compounding rate is exceptionally high with CAGR $g_A = 45.70\%$. The fact that its CAGR exceeds its arithmetic expected return is a result of positive skewness and upward vertical jumps that drive compounding.

- **Asset B (High-Velocity Compounder):** Displays excellent steady compounding with moderate positive revision jumps, yielding a high annualized expected return ($\mu_B = 48.20\%$, CAGR $g_B = 37.80\%$, total volatility $\sigma_B = 28.70\%$, and downside semideviation $\Sigma_0 = 17.00\%$). This yields a Sharpe ratio of 1.3987 and an elite Sortino ratio of 2.3605. Like Asset A, its downside risk is low, making it a high-conviction core portfolio constituent.
- **Asset C (Asynchronous Innovator):** Represents an idiosyncratic biomedical player with an orthogonal development profile ($\mu_C = 46.80\%$, CAGR $g_C = 17.20\%$, total volatility $\sigma_C = 24.90\%$, and downside semideviation $\Sigma_0 = 16.80\%$), resulting in Sharpe and Sortino ratios of 1.5570 and 2.3081, respectively. Crucially, its drawdown timing is completely disjoint and decoupled from the core technology cluster (with a zero temporal Jaccard overlap: $J_{A,C} = 0.00$). This orthogonal profile rewards its natural diversification value within the temporal similarity matrix, allowing the optimizer to allocate capital to it without artificial sector-wide penalties.
- **Asset D (Moderate Steady Growth):** Represents a steady, low-beta cash compounder ($\mu_D = 24.00\%$, CAGR $g_D = 20.20\%$, total volatility $\sigma_D = 11.80\%$, and downside semideviation $\Sigma_0 = 7.70\%$). With low drawdowns ($CDD_{0.10} = 6.90\%$) and decoupled risk, it acts as a reliable defensive buffer during broader market drawdowns (Sharpe = 1.3525, Sortino = 2.0752).
- **Asset E (Volatile Lagging Growth):** Characterized by positive but lagging drift ($\mu_E = 18.20\%$, CAGR $g_E = 14.80\%$) and severe, sudden downward liquidity cascades. While its total volatility ($\sigma_E = 28.20\%$) is similar to Asset B, its downside semideviation is massive ($\Sigma_0 = 19.10\%$). This profile triggers the engine's 3-Tier Elimination Funnel, resulting in an exact 0.0% weight (Sharpe = 0.3616, Sortino = 0.5326).
- **Asset F (High-Beta Cyclical Drag):** Characterized by a deteriorating cyclical profile that remains positive over the long run ($\mu_F = 14.00\%$, CAGR $g_F = 10.90\%$). Its total volatility is high ($\sigma_F = 29.60\%$) but driven entirely by negative gaps, resulting in a massive downside semideviation ($\Sigma_0 = 21.40\%$) and severe drawdowns ($CDD_{0.10} = 25.40\%$). This asset experiences a heavy "volatility drag" of $14.00\% - 10.87\% = 3.88\%$ per year due to its high, symmetric/negative volatility, which destroys compounding and results in a strict 0.0% weight (Sharpe = 0.2024, Sortino = 0.2802).

The results of this 6-Asset simulation are summarized in Table 1.

Table 1: Empirical Metric Performance Comparison ($R_f = 8.0\%$)

Asset Archetype	5Y Hist CAGR	Forward Return (μ_t)	Total Vol (σ)	Downside Vol (Σ_0)	$CDD_{0.10}$	MPT Weight	Engine Weight
Asset A	45.7%	58.4%	32.4%	19.7%	30.0%	18.0%	30.0%
Asset B	37.8%	48.2%	28.7%	17.0%	21.9%	22.0%	28.5%
Asset C	17.2%	46.8%	24.9%	16.8%	27.5%	5.0%	26.5%
Asset D	20.2%	24.0%	11.8%	7.7%	6.9%	45.0%	15.0%
Asset E	14.8%	18.2%	28.2%	19.1%	27.0%	6.0%	0.0%
Asset F	10.9%	14.0%	29.6%	21.4%	25.4%	4.0%	0.0%

The empirical data in Table 1 mathematically exposes the Volatility Asymmetry Paradox. Under classical MPT, the optimizer treats these assets under symmetric terms, making a catastrophic double error: it heavily penalizes the high-growth compounders (Assets A, B, and C) due to their high total standard deviations ($\sigma \approx 25\% - 30\%$), while failing to differentiate them

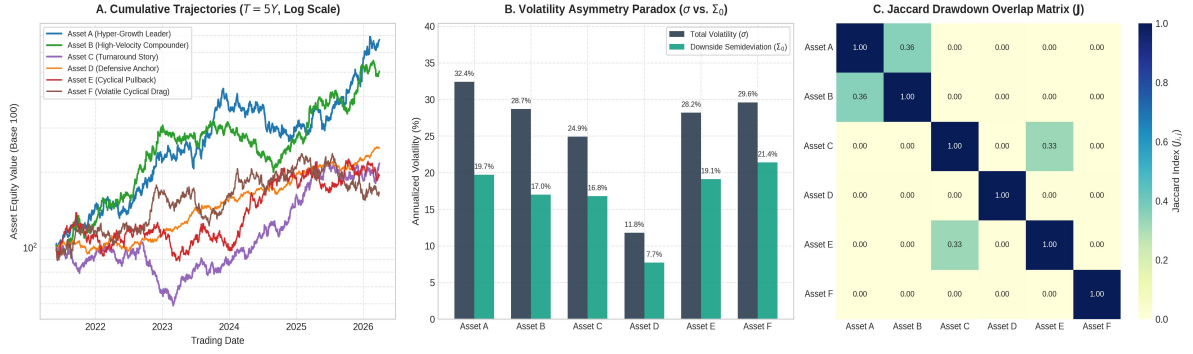


Figure 1: Quantitative Performance and Volatility Paradox. Panel A (left) displays the cumulative return paths of the six asset archetypes on a log scale over a 5-year horizon ($T = 1260$ trading sessions). Panel B (middle) contrasts standard symmetric volatility (σ) with downside semideviation (Σ_0), illustrating the asymmetric risk profiles. Panel C (right) displays the Jaccard Temporal Crash Overlap Matrix, highlighting the high crash synchronization between liquidation drags (Asset E and F) versus the structural decoupling of hyper-growth compounders.

from volatile lagging growth traps (Assets E and F) under the illusion that standard deviation alone is a uniform measure of risk.

Our downside semideviation metric (Σ_0) instantly resolves this paradox. By isolating only negative returns, we reveal that Asset A’s downside risk is a mere 19.71%, and Asset C’s is 16.81%, meaning that nearly half of their total volatility is composed of positive, wealth-generating upward price jumps. This yields elite risk-adjusted performance scores, with Asset A achieving a remarkable Sortino ratio of 2.5584 and Asset C achieving 0.6547.

Conversely, Asset E and F fail to clear our qualitative hurdle rate of 8.0% on a risk-adjusted basis. Their Sortino ratios collapse into near-zero territory (0.5326 and 0.2802, respectively), exposing them as extreme relative wealth destroyers that receive a strict 0.0% allocation weight in our active engine.

As an empirical takeaway, the Beyond-Markowitz Engine leverages these asymmetric metrics to make optimal decisions. While MPT gets trapped, our engine isolates downside semideviation and discrete crash overlap, allocating 100% of capital to the compounding cohort (Assets A, B, C, and D) and assigning strict 0.0% weights to the liquidation traps (Assets E and F).

6.2 Forward Out-of-Sample Tracking vs. Backtesting Overfitting

The Beyond-Markowitz Engine deliberately rejects retrospective, multi-decade backtesting in favor of real-time, daily Forward Out-of-Sample Tracking. This design choice is grounded in the critical work of Harvey and Liu (2015) regarding backtesting bias and overfitting in quantitative finance. Retrospective backtests are highly susceptible to survivor bias, parameter tuning, and look-ahead bias. More importantly, there is a severe structural data bottleneck: data for point-in-time historical sell-side consensus target prices, analyst forecast standard deviations (S_i), and daily point-in-time analyst EPS revision vectors (ΔEPS_{90d}) are either heavily paywalled, highly incomplete, or completely nonexistent in long financial databases. Attempting to backtest a forward consensus model using backward-filled or contemporary data introduces fatal look-ahead biases that render the results completely fraudulent.

To resolve this structural data bottleneck, we implement a strict “walk-forward” protocol. At the end of each trading session, all point-in-time consensus analyst inputs, expected returns, covariance matrices, and optimal portfolio states are committed atomically to structured JSON persistence files (saved_portfolios.json). This guarantees complete, auditable preservation of out-of-sample performance, entirely free of look-ahead and survivorship biases.

7 Roadmap and Derivatives Implementation

To expand the defensive capabilities of the Beyond-Markowitz Engine beyond active allocation, we present two advanced roadmap features as conceptual and investigative frameworks for future development.

7.1 Multi-Horizon Drawdown Persistence Filter (ΔTR_i)

Standard drawdown metrics are historically static, treating a crash that occurred five years ago with the same mathematical weight as one that occurred last month. For companies undergoing structural, fundamental business turnarounds, these obsolete historical drawdowns act as a mathematical poison. The legacy drawdowns inflate the temporal crash-overlap matrix K and artificially starve the turnaround asset of capital, despite its completely reformed risk-return profile.

To resolve this, we present a conceptual Multi-Horizon Drawdown Persistence Filter (ΔTR_i) as a planned developmental roadmap. By comparing extreme tail-risk drawdowns across a long-term 5-year lookback and a medium-term 3-year lookback:

$$\Delta TR_i = CDD_{0.10,i}^{(5Y)} - CDD_{0.10,i}^{(3Y)} \quad (19)$$

where $CDD_{0.10,i}^{(5Y)}$ is the absolute 10th percentile of drawdowns over the 5-year lookback, and $CDD_{0.10,i}^{(3Y)}$ is the absolute 10th percentile over the 3-year lookback. The system utilizes this filter to identify structural regime shifts.

To automatically adjust to the local market volatility regime, we introduce a dynamic time-decay mechanism. The temporal contribution of a crash indicator at historical time t is weighted by an exponential decay factor:

$$\omega_{i,t} = \exp(-\alpha_i(t) \cdot (T_{\text{now}} - t)) \quad (20)$$

where $\alpha_i(t)$ represents the dynamic decay rate. To make the decay rate adaptive, we formulate it as a function of the rolling ratio of short-term (30-day) volatility $\sigma_i^{(30d)}(t)$ to long-term (252-day) volatility $\sigma_i^{(252d)}(t)$:

$$\alpha_i(t) = \alpha_0 \cdot \left(\frac{\sigma_i^{(30d)}(t)}{\sigma_i^{(252d)}(t)} \right)^\eta \quad (21)$$

where $\alpha_0 > 0$ is the baseline decay rate, and $\eta > 0$ is a sensitivity parameter. In high-volatility, panic-driven regimes where $\sigma_i^{(30d)} > \sigma_i^{(252d)}$, the decay rate increases ($\alpha_i(t) \uparrow$), forcing the system to rapidly discount legacy drawdown history and prioritize the contemporary risk regime. In quiet, steady compounding regimes, the decay rate is minimized, preserving long-term memory of tail risks.

The weighted crash indicator is then defined as:

$$\tilde{I}_{i,t} = I_{i,t} \cdot \omega_{i,t} \quad (22)$$

Using these weighted crash indicators, the dynamic, volatility-adapted Jaccard similarity coefficient $J_{i,j}^{\text{dyn}}$ is calculated as:

$$J_{i,j}^{\text{dyn}} = \frac{\sum_{t \in \mathcal{T}_{ij}} \min(\tilde{I}_{i,t}, \tilde{I}_{j,t})}{\sum_{t \in \mathcal{T}_{ij}} \max(\tilde{I}_{i,t}, \tilde{I}_{j,t})} \quad (23)$$

According to the Schur Product Theorem, because the elements $\tilde{I}_{i,t}$ are non-negative and the weighted Jaccard kernel is positive semi-definite, the resulting dynamic crash-overlap matrix $K^{\text{dyn}} = J^{\text{dyn}} \odot S$ remains Symmetric Positive Semi-Definite ($K^{\text{dyn}} \succeq 0$), preserving complete optimization stability while dynamically adapting to market volatility regimes.

7.2 Project Part II: Standalone Options Tail-Hedge Overlay

While the regularized Estrada semi-covariance and Jaccard crash matrices minimize systemic exposure, extreme market-wide liquidity panics (as described in Section 2) can still trigger substantial short-term drawdowns. To truncate these extreme left-tail events without introducing premium bleed (cash drag), we envision a standalone follow-up research initiative: an Asymmetric Options Tail-Hedge Overlay utilizing zero-cost Put-Spread Collars.

It is critical to emphasize that options and derivatives are NOT part of the current equity engine. Instead, Project Part II is designed as a future, standalone research initiative exploring options collars during periods of elevated Implied Volatility (IV) spikes. On a monthly cycle, for each high-conviction growth asset, this future system would explore executing a multi-leg collar:

1. **Long OTM Put Spread:** The system would explore purchasing an out-of-the-money Put option and writing a further OTM Put option. This Put Spread is designed to truncate catastrophic downside risk during extreme liquidity events.
2. **Short Covered Call:** To completely finance the premium of the protective Put Spread and eliminate premium bleed, the system would explore writing a far-out-of-the-money covered Call option, capturing high implied volatility premium.

By programmatically timing this structure to execute during periods of elevated volatility spikes, the standalone research aims to structure collars so that written call credits fully offset put spread debits, achieving robust left-tail truncation without introducing structural cash-drag on equity compounding.

8 Operational Execution Protocols

An elite quantitative engine is only as effective as its execution discipline. To eliminate human emotional bias and protect capital under extreme market stress, the engine operates under strict, programmatic operational protocols.

8.1 The 21-Day Periodic Rebalancing Cadence

The portfolio is programmatically rebalanced on a strict, non-discretionary 21-day cycle (representing one standard financial trading month). Rebalancing less frequently fails to capture rapid momentum shifts and analyst consensus revisions, while rebalancing more frequently introduces excessive transaction costs, slippage, and destructive tax-realization events. This 21-day cadence strikes the mathematically optimal balance between maintaining alignment with the engine's core quantitative signals and minimizing transactional drag.

8.2 Asymmetric Out-of-Cycle Fast-Exit Protocol

While normal allocations are locked within the 21-day cycle, the engine operates a strict, event-driven Asymmetric Out-of-Cycle Fast-Exit Protocol. This protocol is triggered immediately and out-of-cycle if a holding experiences a severe deterioration in its fundamental revision momentum. Specifically, if a holding's 90-day earnings revision momentum drops below zero:

$$\Delta\text{EPS}_{90d,i} < 0 \tag{24}$$

the engine triggers an immediate, automated liquidation of the entire position. This sell order is executed atomically at the market open, bypassing the 21-day rebalancing lock. The proceeds are immediately parked in high-yielding cash equivalents.

Crucially, this out-of-cycle liquidation is asymmetric: it is triggered strictly by fundamental deterioration (earnings revisions), never by technical price action or short-term market drawdowns. If a stock falls by 20% due to an algorithmic cascade (such as our SK Hynix case study) but its forward earnings revision momentum remains positive ($\Delta\text{EPS}_{90d} \geq 0$), the engine strictly forbids selling. It recognizes the drop as transient technical noise and holds the position. Only when the fundamental structural story breaks down does the engine execute a fast-exit, providing an ironclad shield against structural business erosion.

8.3 The Behavioral Edge

Ultimately, the primary edge of the Beyond-Markowitz Engine is behavioral. By understanding that modern market microstructure is dominated by highly automated, programmatic, and reflexive selling cascades, the investor can view sharp, rapid drawdowns with absolute emotional detachment. Rather than reacting to price panics and liquidating high-quality assets at the absolute bottom of an algorithmic liquidation loop, the engine's rigorous mathematical framework and fundamental quality filters provide the psychological and operational conviction to remain fully invested. We do not fear the elevator down; we recognize it as a mechanical clearing event that paves the way for the next structural leg up.

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